

1. Given $X(k)$ is N -pt DFT of $x(n)$, where N is even.
 Define two $\frac{N}{2}$ length sequences: $g(n) = a_1 x(2n) + a_2 x(2n+1)$

and $h(n) = a_3 x(2n) + a_4 x(2n+1)$ where $0 \leq n \leq \frac{N}{2} - 1$

and $a_1, a_4 \neq a_2, a_3$.

If $G(k), H(k) \xleftrightarrow[\frac{N}{2} \text{ pt DFT}]{} g(n), h(n)$.

Find $x(k)$ in terms of $G(k)$ & $H(k)$.

2.
$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

Find $x(n)$

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